Sampling associativity fractions in `cntmosaic`: a slice-within-Gibbs sampler on contingency tables

`cntmosaic` development notes

October 16, 2026

Abstract

This note explains how the function `sample_eta` in `cntmosaic.predict` turns contact counts that are known only by participant group into counts broken down by both participant group and contact group. The method rests on one observation: the two rules every valid answer must obey, *reciprocity* and *sum-to-one*, turn each pair of age cells into a small table with known row and column totals. We put a prior on such tables and sample from it with a Markov chain Monte Carlo (MCMC) method that combines the classic "swap" moves for tables with fixed totals and the slice sampler of Neal (2003). Every step is explained from first principles.

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1 The problem

1.1 What we know and what we want

People are split into K groups, called *strata* (for example $K = 2$: male and female), and into A age groups. We write $k, \ell \in \{1, \dots, K\}$ for strata and $a, b \in \{1, \dots, A\}$ for ages. Throughout, a is the age of the person *reporting* the contact (the participant) and b the age of the person *contacted*.

A partially stratified model, fitted to survey data where only the participant's stratum is recorded, tells us

$$z_{ab}^{k} = \text{total contacts from participants in stratum } k \text{ aged } a \text{ to people aged } b \text{ (any stratum).} \tag{1}$$

We want the finer quantity

$$Z_{ab}^{k\ell} = \text{contacts from stratum } k, \text{ age } a, \text{ to stratum } \ell, \text{ age } b, \tag{2}$$

or equivalently the *associativity fractions*

$$\eta_{ab}^{k\ell} = \frac{Z_{ab}^{k\ell}}{z_{ab}^{k}}, \tag{3}$$

the share of stratum-k participants' contacts (at ages $a \to b$) that go to stratum ℓ. In practice everything below is done separately for each posterior draw of z; we drop the draw index to keep notation light.

1.2 The rules a valid answer must obey

(C1) **Non-negativity:** $Z_{ab}^{k\ell} \geq 0$.

(C2) **Sum-to-one:** the fine counts must add back up to what we know,

$$\sum_{\ell=1}^{K} Z_{ab}^{k\ell} = z_{ab}^{k} \quad \Longleftrightarrow \quad \sum_{\ell=1}^{K} \eta_{ab}^{k\ell} = 1. \tag{4}$$

(C3) **Reciprocity:** every contact has two ends. A contact from (stratum k, age a) to (stratum ℓ, age b) is the same event as a contact from (ℓ, b) to (k, a), so

$$Z_{ab}^{k\ell} = Z_{ba}^{\ell k}. \tag{5}$$

Reciprocity is a standard requirement for contact matrices (Wallinga et al., 2006; Arregui et al., 2018).

Summing (C3) over k and ℓ and using (C2) shows that the known totals must themselves be reciprocal:

$$\sum_{k} z_{ab}^{k} = \sum_{k} z_{ba}^{k}. \tag{6}$$

If (6) fails, no answer satisfies (C1)–(C3). The partially stratified models in `cntmosaic` satisfy it; the code warns and rescales if the input is slightly off.

2 The key observation: one small table per age pair

Rules (C2) and (C3) only ever link a cell (a, b) with its *mirror* cell (b, a). So we can treat each pair of mirror cells on its own.

2.1 Off-diagonal pairs ($a > b$)

Fix $a > b$ and collect the unknown counts at (a, b) in a $K \times K$ table X:

$$X_{k\ell} = Z_{ab}^{k\ell}. \tag{7}$$

- By (C2) at (a, b), the **row sums** of X are known: $\sum_\ell X_{k\ell} = z_{ab}^k =: r_k$.
- By (C3), $X_{k\ell} = Z_{ba}^{\ell k}$, so the counts at the mirror cell are just the **transpose** $X^\top$.
- By (C2) at (b, a), the rows of $X^\top$, i.e. the **column sums** of X, are known: $\sum_k X_{k\ell} = z_{ba}^\ell =: c_\ell$.

Condition (6) says exactly that the row totals and the column totals have the same grand total $N := \sum_k r_k = \sum_\ell c_\ell$.

Definition 1 (Tables with fixed margins)**.** For non-negative vectors $\boldsymbol{r}, \boldsymbol{c} \in \mathbb{R}^K$ with the same total N,

$$\mathcal{T}(\boldsymbol{r}, \boldsymbol{c}) = \Big\{ X \in \mathbb{R}^{K \times K} : X_{k\ell} \geq 0,\ \textstyle\sum_\ell X_{k\ell} = r_k,\ \sum_k X_{k\ell} = c_\ell \Big\}. \tag{8}$$

This set is called a *transportation polytope*.

Proposition 1. *For a pair $a > b$, the counts at (a, b) and (b, a) satisfy (C1)–(C3) if and only if $X \in \mathcal{T}(\boldsymbol{r}, \boldsymbol{c})$ and the counts at (b, a) equal $X^\top$.*

Proof. This is the bullet list above read in both directions. □

So *every* table in $\mathcal{T}(\boldsymbol{r}, \boldsymbol{c})$ is a valid answer, and both constraints hold exactly, by construction. Each entry automatically lies in the classical Fréchet–Hoeffding bounds (Hoeffding, 1940; Fréchet, 1951):

$$\max(0,\ r_k + c_\ell - N) \ \leq\ X_{k\ell} \ \leq\ \min(r_k,\ c_\ell). \tag{9}$$

2.2 Diagonal cells ($a = b$)

When $a = b$ the cell is its own mirror, so (C3) says $X_{k\ell} = X_{\ell k}$: the table is **symmetric**, with row sums (and hence column sums) $r_k = z_{aa}^k$.

2.3 How much freedom is left?

A $K \times K$ table with fixed row and column sums has $(K-1)^2$ free entries; a symmetric one with fixed row sums has $K(K-1)/2$. For $K = 2$ both numbers are 1: *a single number decides everything.*

Example 1 ($K = 2$)**.** Let $x = X_{12}$ (stratum 1 to stratum 2). The margins fix the rest:

$$X = \begin{pmatrix} r_1 - x & x \\ c_1 - r_1 + x & c_2 - x \end{pmatrix}, \qquad x \in \big[\max(0,\ r_1 - c_1),\ \min(r_1,\ c_2)\big]. \tag{10}$$

Check: row 1 sums to r_1, column 1 to c_1, column 2 to c_2, and row 2 to $c_1 + c_2 - r_1 = r_2$ because the totals agree.

3 The prior on tables

3.1 Starting point: one Dirichlet per source and cell

Before imposing reciprocity, a natural prior treats the fractions of each source stratum k at each cell as a Dirichlet vector (see e.g. Ng et al., 2011):

$$\left(\eta_{ab}^{k1},\dots,\eta_{ab}^{kK}\right) \sim \text{Dirichlet}\left(\alpha_{ab}^{k1},\dots,\alpha_{ab}^{kK}\right), \qquad \alpha_{ab}^{k\ell} = \alpha\, q_b^{\ell} + (\text{optional external terms}), \tag{11}$$

where q_b^{ℓ} is the population share of stratum ℓ at age b and $\alpha > 0$ is a precision: larger α pulls the fractions towards the population shares. Recall that the Dirichlet density is $\propto \prod_{\ell} \eta_{\ell}^{\alpha_{\ell}-1}$ on the simplex.

3.2 Imposing reciprocity

At a pair $a > b$ the fractions at both cells are functions of X: $\eta_{ab}^{k\ell} = X_{k\ell}/r_k$ and $\eta_{ba}^{k\ell} = X_{\ell k}/c_k$. Multiplying the Dirichlet densities of both cells and keeping only the tables in $\mathcal{T}(\boldsymbol{r},\boldsymbol{c})$ gives (the r_k and c_k are constants)

$$p(X) \;\propto\; \prod_{k,\ell} X_{k\ell}^{\beta_{k\ell}-1}, \qquad \beta_{k\ell} = \alpha_{ab}^{k\ell} + \alpha_{ba}^{\ell k} - 1, \qquad X \in \mathcal{T}(\boldsymbol{r},\boldsymbol{c}). \tag{12}$$

Each entry $X_{k\ell}$ appears in two places, $\eta_{ab}^{k\ell}$ and $\eta_{ba}^{\ell k}$, so its exponent collects one term from each. For a diagonal cell the same argument gives $p(X) \propto \prod_k X_{kk}^{\alpha_{aa}^{kk}-1} \prod_{k<\ell} X_{k\ell}^{\alpha_{aa}^{k\ell}+\alpha_{aa}^{\ell k}-2}$ on symmetric tables.

Remark (This is not a truncated Dirichlet). A *truncated* Dirichlet restricts the Dirichlet to a region of positive volume (for example a box of bounds). Here we instead *condition* on the equality constraints (C3), a set of lower dimension. The resulting fractions are in general neither Dirichlet nor truncated Dirichlet. Conditioning on a set of probability zero depends on the coordinates used (the Borel–Kolmogorov paradox; Kolmogorov, 1933); we use the table entries, which are linear in the fractions, and the next proposition shows this choice is natural.

Proposition 2 (A Gamma construction)**.** *Let $Y_{k\ell} \sim \text{Gamma}(\beta_{k\ell}, 1)$ be independent, with all $\beta_{k\ell} > 0$. The conditional distribution of the table Y given its row sums $\boldsymbol{r}$ and column sums $\boldsymbol{c}$ has density (12).*

Proof. The joint density of Y is $\prod_{k\ell} y_{k\ell}^{\beta_{k\ell}-1} e^{-y_{k\ell}}/\Gamma(\beta_{k\ell})$. On $\mathcal{T}(\boldsymbol{r},\boldsymbol{c})$ the sum $\sum_{k\ell} y_{k\ell} = N$ is fixed, so $e^{-\sum y_{k\ell}}$ is a constant and only $\prod y_{k\ell}^{\beta_{k\ell}-1}$ remains. □

This mirrors the familiar fact that a Dirichlet vector is a vector of independent Gammas divided by their sum, i.e. conditioned on its total (Ng et al., 2011); here we condition on all row *and* column sums. It is also the continuous analogue of the classical result that independent Poisson counts conditioned on the margins of a table follow a (multivariate) hypergeometric distribution, the setting of Diaconis and Sturmfels (1998).

3.3 When is the prior proper?

Near a face of $\mathcal{T}$ where $X_{k\ell} \to 0$, the density behaves like $X_{k\ell}^{\beta_{k\ell}-1}$, which is integrable only if $\beta_{k\ell} > 0$. With no external terms this reads

$$\beta_{k\ell} = \alpha\left(q_b^{\ell} + q_a^{k}\right) - 1 > 0 \quad\Longleftrightarrow\quad \alpha > \frac{1}{q_b^{\ell} + q_a^{k}}. \tag{13}$$

With two strata of roughly equal size the shares are near $1/2$, so we need α somewhat above 1; this is why the default is $\alpha = 2$. The code checks (13) and raises an error that reports the smallest admissible α. (For $K = 2$ one checks directly from Example 1 that the condition is also necessary.)

4 Sampling: Gibbs moves along margin-preserving directions

We cannot draw from (12) directly: its normalising constant has no closed form. We therefore use MCMC: build a Markov chain on $\mathcal{T}$ whose long-run distribution is (12) (Robert and Casella, 2004).

4.1 Moves that keep the margins

Write $E_{k\ell}$ for the table with a 1 in position (k, ℓ) and zeros elsewhere. For rows $k \neq k'$ and columns $\ell \neq \ell'$ define the *swap direction*

$$M = E_{k\ell} + E_{k'\ell'} - E_{k\ell'} - E_{k'\ell} \qquad \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix} \text{ on rows } k, k' \text{ and columns } \ell, \ell'. \tag{14}$$

Every row and column of M sums to zero, so $X + dM$ has the same margins as X for every number d. These 2×2 moves are the classical tool for exploring tables with fixed margins (Diaconis and Gangolli, 1995; Diaconis and Sturmfels, 1998). For symmetric (diagonal) tables we use $M = E_{k\ell} + E_{\ell k} - E_{kk} - E_{\ell\ell}$, which keeps both symmetry and the row sums.

Non-negativity limits how far we can move: $X + dM \in \mathcal{T}$ exactly when

$$d \in [d_-, d_+], \qquad d_- = -\min(X_{k\ell},\, X_{k'\ell'}), \qquad d_+ = \min(X_{k\ell'},\, X_{k'\ell}). \tag{15}$$

4.2 The one-dimensional target

Along the move, the density (12) becomes a function of d only:

$$g(d) \;\propto\; (X_{k\ell} + d)^{\beta_{k\ell}-1}\,(X_{k'\ell'} + d)^{\beta_{k'\ell'}-1}\,(X_{k\ell'} - d)^{\beta_{k\ell'}-1}\,(X_{k'\ell} - d)^{\beta_{k'\ell}-1}, \qquad d \in [d_-, d_+]. \tag{16}$$

Drawing d from g and setting $X \leftarrow X + dM$ is a *Gibbs update along a direction* (Gelfand and Smith, 1990), closely related to the hit-and-run sampler of Smith (1984). Each such update leaves (12) invariant, and so does any fixed sequence of them (Tierney, 1994, Section 2.4). One *sweep* applies every swap direction once; the swap directions span every direction in which a table in $\mathcal{T}$ can move, so repeated sweeps can explore all of $\mathcal{T}$.

For $K = 2$ there is exactly one direction, and d is simply the change in $x = X_{12}$ of Example 1.

5 The slice sampler

5.1 Why not sample g directly?

g is a product of up to four powers of linear functions. It is not a standard distribution, and neither its cumulative distribution function nor its inverse has a closed form, so we cannot use the inverse-CDF method. Common alternatives have drawbacks here:

- **Random-walk Metropolis** needs a step size, and the right size varies enormously across the thousands of age pairs and posterior draws we update at once.
- **Rejection sampling** needs an envelope function that lies above g. When an exponent $\beta - 1$ is between -1 and 0, g is infinite at an end of the interval (though still integrable), which makes envelopes awkward.

The slice sampler of Neal (2003) avoids both problems: it has no step size to tune, copes with infinite spikes at the ends, needs g only up to a constant, and leaves g exactly invariant.

5.2 The idea

Sampling from a density is the same as picking a point uniformly from the region under its graph, $\{(d, u) : 0 < u < g(d)\}$, and keeping the d coordinate (Neal, 2003; Damien et al., 1999). The slice sampler does this by alternating two easy steps from the current point d_0:

1. **Pick a height.** Draw $u \sim \text{Uniform}(0, g(d_0))$.
2. **Pick a point on the slice.** Draw the new d uniformly from the *slice* $S = \{d : g(d) > u\}$.

Both steps are Gibbs updates for the uniform distribution under the graph, so the pair (d, u) keeps that distribution, and d keeps distribution g.

For numerical safety we work with logarithms: step 1 is equivalent to $y = \log g(d_0) - \varepsilon$ with $\varepsilon \sim \text{Exponential}(1)$, and the slice is $S = \{d : \log g(d) > y\}$.

5.3 Finding a point on the slice: shrinkage

We usually cannot write S down, but we can test whether a given d is in it. Neal's *shrinkage* procedure (Neal, 2003, Section 4.2) starts from an interval $[L, R]$ that contains the whole slice and repeats:

1. propose $d^\star \sim \text{Uniform}(L, R)$;
2. if $d^\star \in S$, accept it and stop;
3. otherwise shrink the interval towards the current point: if $d^\star < d_0$ set $L = d^\star$, else set $R = d^\star$.

Because the current point d_0 is always in S and always stays inside $[L, R]$, the loop ends. Neal proves that this procedure leaves the uniform distribution on S invariant, even when S is not a single interval (Neal, 2003, Sections 4.2–4.3); this matters here because g can be U-shaped, giving a slice made of two pieces.

Our situation is the easiest one Neal describes: g lives on the bounded interval $[d_-, d_+]$, so we start with $[L, R] = [d_-, d_+]$, which certainly contains S. No "stepping-out" phase is needed.

5.4 The full algorithm

Algorithm 1 One slice update of a table X along direction M (current step $d_0 = 0$)

$d_- \leftarrow -\min(X_{k\ell}, X_{k'\ell'})$, $d_+ \leftarrow \min(X_{k\ell'}, X_{k'\ell})$
if $d_- = d_+$ **then return** X ▷ no room to move
end if
$y \leftarrow \log g(0) - \varepsilon$, $\varepsilon \sim \text{Exponential}(1)$
$L \leftarrow d_-$, $R \leftarrow d_+$
loop
    $d^\star \sim \text{Uniform}(L, R)$
    if $\log g(d^\star) > y$ **then**
        return $X + d^\star M$
    else if $d^\star < 0$ **then**
        $L \leftarrow d^\star$
    else
        $R \leftarrow d^\star$
    end if
end loop

Algorithm 2 `sample_eta`: one table per age pair and posterior draw

for each posterior draw and each age pair $a \geq b$ **do**
  $\boldsymbol{r} \leftarrow (z^k_{ab})_k$, $\boldsymbol{c} \leftarrow (z^\ell_{ba})_\ell$, $N \leftarrow \sum_k r_k$
  $X \leftarrow \boldsymbol{r}\boldsymbol{c}^\top / N$ ▷ independence table: strictly inside $\mathcal{T}$
end for
for sweep $= 1, \ldots, n_{\text{sweeps}}$ **do**
  for each swap direction M (symmetric directions on the diagonal) **do**
    update every table with Algorithm 1
  end for
end for
$\eta^{k\ell}_{ab} \leftarrow X_{k\ell}/r_k$, $\eta^{k\ell}_{ba} \leftarrow X_{\ell k}/c_k$

6 Implementation notes

- **Starting point.** The independence table $\boldsymbol{r}\boldsymbol{c}^\top/N$ has the right margins and positive entries whenever the margins are positive, so it is a valid starting point. MCMC draws are only exact in the limit; after a finite number of sweeps they still depend slightly on where the chain started.
- **Number of sweeps.** The default is $n_{\text{sweeps}} = 10$. In a test cell with an exactly computable mean, 1–3 sweeps were visibly biased towards the starting point, while 5 or more sweeps matched the exact mean.
- **Vectorisation.** All tables (every posterior draw and age pair) run as independent chains, updated together with NumPy. Almost every chain accepts its first proposal (on average 1.04 proposals per update), so after the first proposal the code only keeps working on the few chains that have not yet accepted. This gave a 6× speed-up without changing the algorithm.
- **Zero margins.** If a source has no contacts at a cell ($r_k = 0$), its row is zero, every move involving it has $d_- = d_+ = 0$, and its fractions are reported as 0.

7 Checks

The unit tests in `cntmosaic/predict/tests/test_predict.py` check:

- sum-to-one (C2) and reciprocity (C3) hold to machine precision (around 10^{-16}), for $K = 2$ and $K = 3$;
- every fraction lies within its Fréchet–Hoeffding bounds;
- for $K = 2$, the sampled mean of x in Example 1 agrees, within four standard errors, with the exact mean of (16) computed by numerical integration;
- an improper prior (condition (13) violated) raises an error, and non-reciprocal input totals trigger a warning.

8 Summary and limitations

- Reciprocity and sum-to-one turn each age pair into a table with known row and column totals (Proposition 1); any such table is a valid answer.

- The prior is the Dirichlet prior of each cell, conditioned on reciprocity; equivalently, independent Gamma entries conditioned on the margins (Proposition 2). It is proper when condition (13) holds.
- We sample it with Gibbs updates along margin-preserving swap directions (Diaconis and Sturmfels, 1998), each drawn exactly with a shrinkage slice sampler (Neal, 2003).

Limitations. (i) The draws are approximate for a finite number of sweeps. (ii) The conditioned prior depends on the choice of coordinates (see the Remark in Section 3); we use the table entries. (iii) We are not aware of a published source that uses exactly this construction for contact matrices; the components above are standard, and the closest analogue is the Diaconis–Sturmfels approach for discrete tables.

References

Arregui, S., Aleta, A., Sanz, J. and Moreno, Y. (2018). Projecting social contact matrices to different demographic structures. *PLoS Computational Biology*, 14(12), e1006638.

Damien, P., Wakefield, J. and Walker, S. (1999). Gibbs sampling for Bayesian non-conjugate and hierarchical models by using auxiliary variables. *Journal of the Royal Statistical Society: Series B*, 61(2), 331–344.

Diaconis, P. and Gangolli, A. (1995). Rectangular arrays with fixed margins. In *Discrete Probability and Algorithms*, IMA Volumes in Mathematics and its Applications, vol. 72, 15–41. Springer.

Diaconis, P. and Sturmfels, B. (1998). Algebraic algorithms for sampling from conditional distributions. *The Annals of Statistics*, 26(1), 363–397.

Fréchet, M. (1951). Sur les tableaux de corrélation dont les marges sont données. *Annales de l'Université de Lyon, Section A*, 14, 53–77.

Gelfand, A. E. and Smith, A. F. M. (1990). Sampling-based approaches to calculating marginal densities. *Journal of the American Statistical Association*, 85(410), 398–409.

Hoeffding, W. (1940). Masstabinvariante Korrelationstheorie. *Schriften des Mathematischen Instituts und des Instituts für Angewandte Mathematik der Universität Berlin*, 5, 181–233.

Kolmogorov, A. N. (1933). *Grundbegriffe der Wahrscheinlichkeitsrechnung*. Springer, Berlin.

Neal, R. M. (2003). Slice sampling. *The Annals of Statistics*, 31(3), 705–767.

Ng, K. W., Tian, G.-L. and Tang, M.-L. (2011). *Dirichlet and Related Distributions: Theory, Methods and Applications*. Wiley.

Robert, C. P. and Casella, G. (2004). *Monte Carlo Statistical Methods*, 2nd edition. Springer.

Smith, R. L. (1984). Efficient Monte Carlo procedures for generating points uniformly distributed over bounded regions. *Operations Research*, 32(6), 1296–1308.

Tierney, L. (1994). Markov chains for exploring posterior distributions. *The Annals of Statistics*, 22(4), 1701–1762.

Wallinga, J., Teunis, P. and Kretzschmar, M. (2006). Using data on social contacts to estimate age-specific transmission parameters for respiratory-spread infectious agents. *American Journal of Epidemiology*, 164(10), 936–944.